Software Requirements Specification for CEWS System: Data Separability Measurement

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Contents

Revision History

Date	Version	Notes
2023-01-24	1.0	Creation

1 Reference Material

This section records information for easy reference.

1.1 Table of Units

Throughout this document no units are used.

1.2 Table of Symbols

The table that follows summarizes the symbols used in this document along with their units. The choice of symbols was made to be consistent with the heat transfer literature and with existing documentation for solar water heating systems. The symbols are listed in alphabetical order.

Symbol	Unit	Description
J_N	N/A	Normalized Multiclass Cut Edge Weight Measurement
$I(...)$	N/A	Indicator Function

1.3 Abbreviations and Acronyms

Symbol	Description
A	Assumption
DD	Data Definition
GD	General Definition
GS	Goal Statement
IM	Instance Model
LC	Likely Change
PS	Physical System Description
R	Requirement
RNG	Relative Neighbourhood Graph
SRS	Software Requirements Specification
CEWS	Cut Edge Weight Statistic
T	Theoretical Model

1.4 Mathematical Notation

This document uses notation used in (1).

2 Introduction

Classification is a common machine learning problem and there are many techniques for solving classification problems. Unfortunately, not all classification problems are easy and it can be difficult to explain why classifiers preform poorly on certain datasets. Data complexity has been defined as a characteristic of the data that relates to the difficulty of the classification problem (2). To further refine this definition, we consider data complexity to be represented by aspects of the data that introduce difficulty for the classifiers.

One of these aspects is separability, an intrinsic characteristic that describes how the data points from different classes mix together (3). A classic measurement for separability is Fraction of Points on Class Boundary. Based on the work of Friedman and Rafsky (4) and documented by Basu and Ho(5), this measurement constructs a minimum spanning tree over the dataset then calculates the number of points with a connection to another class over all points.

The Cut Edge Weight Statistic measurement (1) is a variation of the Fraction of Points on Class Boundary, proposed by Zighed et al. The algorithm for the normalized measurement begins by constructing a Toussaint's Relative Neighbourhood Graph (6) over the entire dataset then computing the number of edges which connect different classes over the total number of edges. This measurement is a viable way of measuring separability because it has a high coefficient of determination (R^2) with the mean error rate of several classifiers.

The introduction contains the purpose of this document, the scope of requirements, characteristics of the indented reader and the organization of this complete document.

2.1 Purpose of Document

The purpose of this document is to provide information about the design and functionality of CEWS System to the Intended Reader (Section 2.3).

2.2 Scope of Requirements

The domain of this problem is restricted to generating the normalized multiclass CEWS Measurement (1). Notable exclusions include the calculation of a p-value, variations of the calculation using different weighing parameters and different graph generation algorithms. The scope excludes asymptotic cost limitations and efficiency requirements.

2.3 Characteristics of Intended Reader

The reader should have a good understanding of classification (graduate level) and a base understanding of data analysis, machine learning and statistics.

2.4 Organization of Document

This document includes a general system description, a specific system description, a requirements section, a section on likely changes, a section on unlikely changes and a section on traceability.

3 General System Description

This section provides general information about the system. It identifies the interfaces between the system and its environment, describes the user characteristics and lists the system constraints.

3.1 System Context

The CEWS System will exist as a standalone product or as a module. The system context is illustrated in Figure 1.

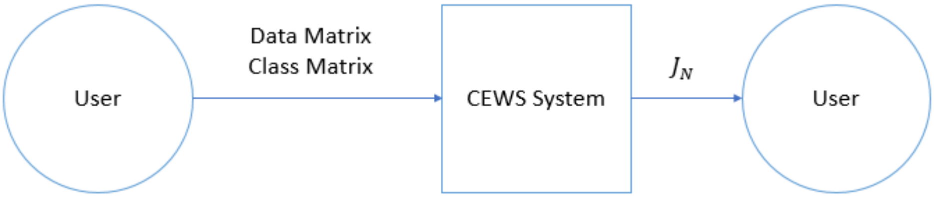


Figure 1: System Context

- User Responsibilities:
  - Provide the data.
  - Correctly format the input data.
  - Resource management including supplying the system with enough resources to support the size of their dataset.
  - Understand the problem.
  - Preform any data pre-processing required for the relevant classification problem such as transformations, feature extraction, etc.
  - Understand the CEWS algorithm.
  - Understand the limitations of numerical computation.
- CEWS System Responsibilities:
  - Read input data.
  - Detect data type mismatch or if input data violates any constraints.
  - Preform the analysis.
  - Preform basic error checking on the output.
  - Return the output value.

3.2 User Characteristics

The end user of CEWS System should have a good understanding of classification and machine learning, preferably with an undergraduate degree in software.

3.3 System Constraints

At this time, there are no known constraints for this system..

4 Specific System Description

This section first presents the problem description, which gives a high-level view of the problem to be solved. This is followed by the solution characteristics specification, which presents the assumptions, theories, definitions and finally the instance models.

4.1 Problem Description

CEWS System is intended to solve the calculation of the normalized multiclass CEWS measurement for a dataset.

4.1.1 Terminology and Definitions

This subsection provides a list of terms that are used in the subsequent sections and their meaning, with the purpose of reducing ambiguity and making it easier to correctly understand the requirements:

- Data point/Example : An identifiable element in a data set which has features associated with it.
- Data set/Sample: A set of examples from a population.
- Feature : An individual measurable property of an example.
- Class Label : Represents the class an example belongs to.
- Classification: the problem of identifying which of a set of categories an example belongs to.
- Data complexity: aspects of the data that classifiers have difficultly with. (2)
- Separability: an intrinsic characteristic of a dataset and describes how the data points from different classes mix together. (3)

4.1.2 Physical System Description

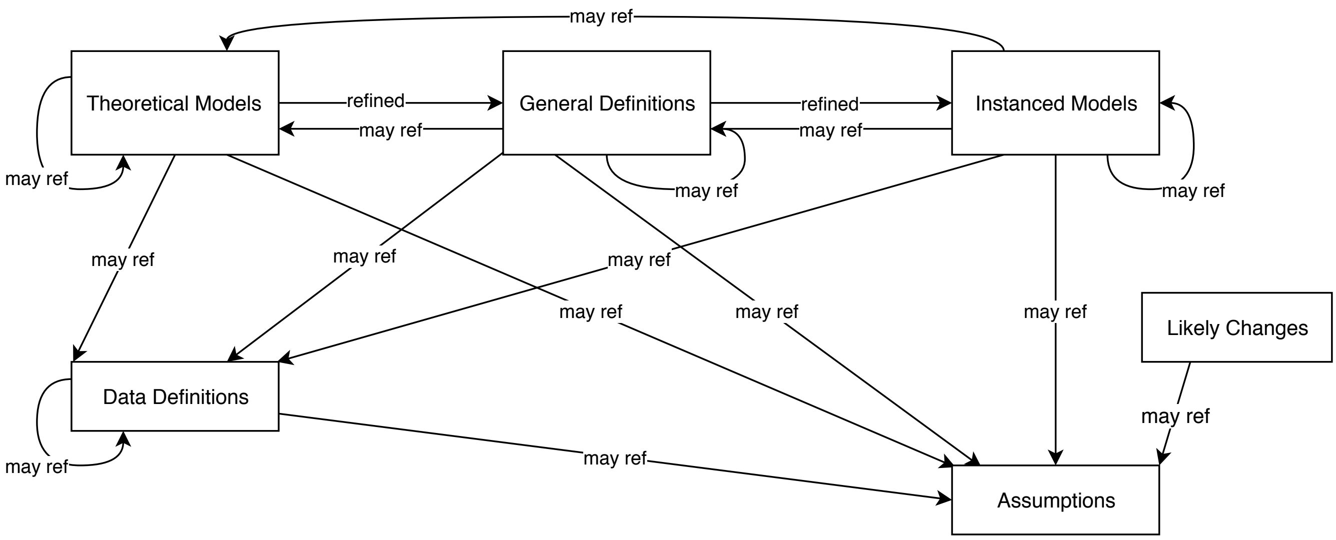
There is no physical system.

4.1.3 Goal Statements

Given the dataset, the goal statements are:

GS1: Calculate the normalized multiclass CEWS measurement.

4.2 Solution Characteristics Specification



The instance models that govern CEWS System are presented in Subsection 4.2.5. The information to understand the meaning of the instance models and their derivation is also presented, so that the instance models can be verified.

4.2.1 Assumptions

This section simplifies the original problem and helps in developing the theoretical model by filling in the missing information for the physical system. The numbers given in the square brackets refer to the theoretical model [T], general definition [GD], data definition [DD], instance model [IM], or likely change [LC], in which the respective assumption is used.

A1: All rational numbers can be represented in a base 2 computer system. Operations preformed on rational numbers in a computer system is exact. Any computation rounding error is negligible and does not need to be taken into account.

A2: Class labels are represented as whole numbers. All examples from one class share the same class label.

A3: Each example has the same number of features. No features are null.

A4: The user does not want to use weights for the edge calculation.

A5: The user wants to use a relative neighbourhood graph as the base graph.

A6: The user doesn't want the statistic p-value.

4.2.2 Theoretical Models

No theoretical models are used.

4.2.3 General Definitions

This section collects and equations that will be used in building the instance models.

Number	GD1
Label	**Toussaint's Relative Neighbourhood Graph**
SI Units	N/A
Equation	$E_{RNG} : d(a,b) \leq \max(d(a,c), d(b,c)) \forall c, c \neq a, b$
Description	RNG is type of graph constructed V is a set of point in real space with m dimensions. G_{RNG} is a graph with vertices set V and the set of edges of points (a,b) that met the criteria of E_{RNG}. $d(u,v)$ is the Euclidean distance between points u and v given by DD1.
Source	(1)
Ref. By	IM3 IM2

4.2.4 Data Definitions

This section collects and defines all the data needed to build the instance models. The dimension of each quantity is also given.

Number	DD1
Label	**Euclidean distance**
Symbol	$d(a,b)$
SI Units	N/A
Equation	$d(a,b) = \sqrt{\sum_{i=1}^{n}(a_i - b_i)^2}$
Description	$d(a,b)$ is the Euclidiean distance between points a and b in Euclidean space with n dimensions. a_i and b_i are Euclidean vectors from the origin.
Sources	N/A
Ref. By	GD1

4.2.5 Instance Models

This section transforms the problem defined in Section 4.1 into one which is expressed in mathematical terms. It uses concrete symbols defined in Section 4.2.4 to replace the abstract symbols in the models identified in Sections 4.2.2 and 4.2.3.

The goals are solved by .

Number	IM1
Label	**Multiclass Normalized Cut Edge Weight Measurement**
Input	I from IM3 and J from IM2.
Output	J_N, $0 \leq J_N \leq 1$, such that $J_N = \frac{J}{I+J}$
Description	The above equation gives the formula for calculating the the multiclass normalized cut edge weight measurement.
Sources	(1)
Ref. By	None

Number	IM2
Label	**Multiclass Sum of Cut Edges**
Input	The input data matrix, D, is of size $n \times m$ is formatted as: $$D_{n\times m} = \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1m} \\ d_{21} & d_{22} & \cdots & d_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & \cdots & d_{nm} \end{bmatrix}$$ The class label matrix, C, if of size n and is formatted as: $$C_{=} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$ The input is constrained based on 4.2.6.
Output	J, $0 \leq J \leq n^2$, such that $J = \sum_{r=1}^{k-1} \sum_{s=r+1}^{k} J_{r,s} = \frac{1}{2} \sum_2 Z_{ij}$
Description	The above equation gives the formula for calculating the multiclass is the sum of cut edges. k is the number of classes. $J_{r,s}$ is the sum of edges linking a vertex of class r and a vertex of class s based on GD1. Z_{ij} is $I(c_i = c_j)$, a boolean variable indicating different classes (one if the vertices i and j have different and zero if not).
Sources	(1) (7)
Ref. By	IM1

Number	IM3
Label	**Multiclass Sum of Non Cut Edges**
Input	The input data matrix, D, is of size $n \times m$ is formatted as: $$D_{n\times m} = \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1m} \\ d_{21} & d_{22} & \cdots & d_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & \cdots & d_{nm} \end{bmatrix}$$ The class label matrix, C, if of size n and is formatted as: $$C_= \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$ The input is constrained based on 4.2.6.
Output	I, $0 \leq I \leq n^2$, such that $I = \sum_{r=1}^{k} I_r = \frac{1}{2} \sum_2 T_{ij}$
Description	The above equation gives the formula for calculating the the multiclass sum of non cut edges. k is the number of classes. I_r is the sum of edges linking two vertices of the same class for class r based on GD1. T_{ij} is $I(c_i = c_j)$, a boolean variable indicating same class (one if the vertices i and j have the classes are the same and zero if not).
Sources	(1) (7)
Ref. By	IM1

4.2.6 Input Data Constraints

Table 1 shows the data constraints on the input output variables. The column for physical constraints gives the physical limitations on the range of values that can be taken by the variable. The column for software constraints restricts the range of inputs to reasonable values. The software constraints will be helpful in the design stage for picking suitable algorithms. The constraints are conservative, to give the user of the model the flexibility to experiment with unusual situations. The column of typical values is intended to provide a feel for a common scenario. The uncertainty column provides an estimate of the confidence with which the physical quantities can be measured. This information would be part of the input if one were performing an uncertainty quantification exercise.

The input data matrix, D, is of size $n \times m$, must contain rational values. n is the number of examples. m is the number of features. The size of n and m are limited based on the system hardware but not by CEWS itself. It is the user's responsibility to insure that their system has the relevant requirements to support their dataset size. It is the user's responsibility to set c_i to the class of d_i.

Table 1: Input Variables

Var	**Constraints**	**Software Constraints**	**Typical Value**	**Uncertainty**
d_{ij}	$d_{ij} \in \mathbb{R}$	$d_{ij} \in \mathbb{Q}, F_{\min} \leq d_{ij} \leq F_{max}$	N/A	N/A
c_i	$c_i \in \mathbb{N}$	$W_{\min} \leq c_i \leq W_{\max}$	N/A	N/A
n	$n \in \mathbb{N}, n > 0$	N/A	N/A	N/A
m	$m \in \mathbb{N}, m > 0$	N/A	N/A	N/A

Table 2: Specification Parameter Values

Var	**Value**
$F_{\min}$	-(2 -2^{23})2^{127}
$F_{\max}$	(2-2^{23})2^{127}
$W_{\min}$	0
$W_{\max}$	$2^8 - 1$

4.2.7 Properties of a Correct Solution

A correct solution must exhibit the following behaviour

Table 3: Output Variables

Var	**Physical Constraints**
J_N	$0 \leq J_N \leq 1$ (by A1)

5 Requirements

This section provides the functional requirements, the business tasks that the software is expected to complete, and the nonfunctional requirements, the qualities that the software is expected to exhibit.

5.1 Functional Requirements

R1: CEWS System will read the input data.

- Any constraint in 4.2.6 is violated then CEWS System will generate an appropriate error message.
- The input matrices are formatted as array-like objects.

R2: After reading the input data, CEWS System will preform the calculation in IM1.

R3: The CEWS System should generate an error if the calculation result exceeds the constraints in 4.2.7.

R4: The CEWS System will generate an output.

- In case an error occurs, an error should be raised by the program.
- In case no error occurs, no error should be raised and J_N will be returned as a float type.

5.2 Nonfunctional Requirements

NFR1: **Accuracy** As per A1, any calculation error should be negligible. The accuracy of the computed solutions should meet the level needed for data analysis. The absolute error should be less than 0.01.

NFR2: **Usability** A program should be able to call CEWS System by calling a function of the form CEWS(D, C) and CEWS System should return the output as described in R4.

NFR3: **Maintainability** The program code should be redable and clear.

NFR4: **Portability** The software should run on the Windows 11 operating system.

6 Likely Changes

LC1: The addition of user options such as option to weigh the edge calculations: IM3 and IM2. Related to: A4.

LC2: The addition of user options such as option to change the algorithm for graph construction: IM3 and IM2. Related to: A5.

LC3: Changes to the output format. Related to: A6.

7 Unlikely Changes

LC4: The input dataset format changes.

8 Traceability Matrices and Graphs

The purpose of the traceability matrices is to provide easy references on what has to be additionally modified if a certain component is changed. Every time a component is changed, the items in the column of that component that are marked with an "X" may have to be modified as well. Table 5 shows the dependencies of theoretical models, general definitions, data definitions, and instance models with each other. Table 6 shows the dependencies of instance models, requirements, and data constraints on each other. Table 4 shows the dependencies of theoretical models, general definitions, data definitions, instance models, and likely changes on the assumptions.

	A1	A2	A3	A4	A5	A6
GD1	X		X			
DD1	X		X			
IM2	X	X	X			
IM3	X	X	X			
IM1	X	X	X			
LC1				X		
LC3					X	
LC4						X

Table 4: Traceability Matrix Showing the Connections Between Assumptions and Other Items

	GD1	DD1	IM2	IM3	IM1
GD1		x	x	x	
DD1	x				
IM2	x				x
IM3	x				x
IM1			x	x	

Table 5: Traceability Matrix Showing the Connections Between Items of Different Sections

	IM2	IM3	IM1
IM2			X
IM3			X
IM1	x	x	
R1	X	X	
R2	X	X	X
R3			X
R4			X

Table 6: Traceability Matrix Showing the Connections Between Requirements and Instance Models

The purpose of the traceability graphs is also to provide easy references on what has to be additionally modified if a certain component is changed. The arrows in the graphs represent dependencies. The component at the tail of an arrow is depended on by the component at the head of that arrow. Therefore, if a component is changed, the components that it points to should also be changed. Figure **??** shows the dependencies of theoretical models, general definitions, data definitions, instance models, likely changes, and assumptions on each other. Figure **??** shows the dependencies of instance models, requirements, and data constraints on each other.

References

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